

simplified here by making them all one single step, or finite standard deviation. But the problem is that some fractal payoffs may have finite variance, but still not take us there rapidly. See Bouchaud and Potters (2003).

Lognormal: There is an intermediate variety that is called the lognormal, emphasized by one Gibrat (see Sutton [1997]) early in the twentieth century as an attempt to explain the distribution of wealth. In this framework, it is not quite that the wealthy get wealthier, in a pure preferential attachment situation, but that if your wealth is at 100 you will vary by 1, but when your wealth is at 1,000, you will vary by 10. The relative changes in your wealth are Gaussian. So the lognormal superficially resembles the fractal, in the sense that it may tolerate some large deviations, but it is dangerous because these rapidly taper off at the end. The introduction of the lognormal was a very bad compromise, but a way to conceal the flaws of the Gaussian.

Extinctions: Sterelny (2001). For extinctions from abrupt fractures, see Courtillot (1995) and Courtillot and Gaudemer (1996). Jumps: Eldredge and Gould.

FRACTALS, POWER LAWS, and SCALE-FREE DISTRIBUTIONS

Definition: Technically, $P > x = K x^{-\alpha}$ where α is supposed to be the power-law exponent. It is said to be scale free, in the sense that it does not have a characteristic scale: relative deviation of $\frac{P > x}{P > nx}$ does not depend on x , but on n —for x “large enough.” Now, in the other class of distribution, the one that I can intuitively describe as nonscalable, with the typical shape $p(x) = \text{Exp}[-a x]$, the scale will be a .

Problem of “how large”: Now the problem that is usually misunderstood. This scalability might stop somewhere, but I do not know where, so I might consider it infinite. The statements *very large* and *I don’t know how large* and *infinitely large* are epistemologically substitutable. There might be a point at which the distributions flip. This will show once we look at them more graphically.

$\text{Log } P > x = -\alpha \text{ Log } X + C^t$ for a scalable. When we do a log-log plot (i.e., plot $P > x$ and x on a logarithmic scale), as in [Figures 15](#) and [16](#), we should see a straight line.

Fractals and power laws: Mandelbrot (1975, 1982). Schroeder (1991) is imperative. John Chipman’s unpublished manuscript *The Paretian Heritage* (Chipman [2006]) is the best review piece I’ve seen. See also